

Bachelor in Computer Applications

Course Title: **Computer Graphics**

Code No: Semester: **5**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. Write the algorithm for Bresenham's line drawing? Digitize a line with end points A(6, 9) and B(10,5) using Bresenham's line drawing algorithm.
2. How can polygons be clipped? Describe Gouraud Shading model with necessary derivations.
3. Differentiate between parallel and perspective projection. Perform the rotation of triangle with vertices A(5,7), B(12,85) and C(5,3) counterclockwise 30 degree about the pivot point (5,3).
4. Explain motion capture with an example.
5. Explain architecture of random scan display systems with suitable diagram.
6. Explain Cohen-Sutherland Line Clipping algorithm with example.
7. Explain about parametric curve. Describe the properties of Bezier curve.
8. Reflect a line segment having end points (9,3) and (12,10) about a line $X = 7$. Draw initial and final result graph as well.
9. Explain ambient light, diffuse reflection and specular reflection with examples.
10. Explain the wireframe representation of 3D objects.
11. How does a polygon can be created in OpenGL? Illustrate with an example.
12. Write short notesa. Z-Bufferb. Polygon Table